

Homework Assignment 5

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1. (2D.2) Prove that there exists a bounded set $A \subset \mathbb{R}$ such that $|F| \leq |A| - 1$ for every closed set $F \subset A$.

Proof. Choose a bounded set A which is not Lebesgue measurable. Then there exists an $\varepsilon > 0$ such that given any closed set F contained in A , we have $|A \setminus F| \geq \varepsilon$. Now consider $\frac{1}{\varepsilon}A$. Each closed subset of $\frac{1}{\varepsilon}A$ is of the form $\frac{1}{\varepsilon}F$ for some closed $F \subset A$. Since $\frac{1}{\varepsilon}F$ and F are Borel,

$$\begin{aligned} \left| \frac{1}{\varepsilon}A \setminus \frac{1}{\varepsilon}F \right| &= \left| \frac{1}{\varepsilon}A \right| - \left| \frac{1}{\varepsilon}F \right| \\ &= \frac{1}{\varepsilon} (|A| - |F|) \\ &= \frac{1}{\varepsilon} |A \setminus F|. \end{aligned}$$

We can now conclude that

$$\begin{aligned} \left| \frac{1}{\varepsilon}F \right| &= \left| \frac{1}{\varepsilon}A \right| - \left| \frac{1}{\varepsilon}A \setminus \frac{1}{\varepsilon}F \right| \\ &= \left| \frac{1}{\varepsilon}A \right| - \frac{1}{\varepsilon} |A \setminus F| \\ &\leq \left| \frac{1}{\varepsilon}A \right| - 1, \end{aligned}$$

so $\frac{1}{\varepsilon}A$ is our desired set. □

2. (2D.5) Prove that if $A \subset \mathbb{R}$ is Lebesgue measurable, then there exists an increasing sequence $F_1 \subset F_2 \subset \dots$ of closed sets contained in A such that

$$\left| A \setminus \bigcup_{k=1}^{\infty} F_k \right| = 0.$$

Proof. By Theorem 2.71, we can choose closed sets $G_1, G_2, \dots$ contained in A such that $|A \setminus \bigcup_{k=1}^{\infty} G_k| = 0$. We want the sequence to be increasing, so define $F_n = \bigcup_{k=1}^n G_k$. Then $F_n \subset F_{n+1}$ for all n , and each F_n is closed since it is a finite union of closed sets. Since $\bigcup_{k=1}^{\infty} F_k = \bigcup_{k=1}^{\infty} G_k$, we have

$$\left| A \setminus \bigcup_{k=1}^{\infty} F_k \right| = \left| A \setminus \bigcup_{k=1}^{\infty} G_k \right| = 0.$$

□

3. (2D.7) Prove that if $A \subset \mathbb{R}$ is Lebesgue measurable, then there exists a decreasing sequence $G_1 \supset G_2 \supset \dots$ of open sets containing A such that

$$\left| \left(\bigcap_{k=1}^{\infty} G_k \right) \setminus A \right| = 0.$$

Proof. By Theorem 2.71, there exist open sets $F_1, F_2, \dots$ containing A such that

$$\left| \left(\bigcap_{k=1}^{\infty} F_k \right) \setminus A \right| = 0.$$

Similar to the proof to 2D.5, we can now define $G_1, G_2, \dots$ by $G_n = \bigcap_{k=1}^n F_k$. Then each G_n is open and contains A , and $\{G_k\}$ is a decreasing sequence, so

$$\left| \left(\bigcap_{k=1}^{\infty} G_k \right) \setminus A \right| = \left| \left(\bigcap_{k=1}^{\infty} F_k \right) \setminus A \right| = 0.$$

□

4. (2D.9) Prove that the collection of Lebesgue measurable subsets of $\mathbb{R}$ is dilation invariant. More precisely, prove that if $A \subset \mathbb{R}$ is Lebesgue measurable and $t \in \mathbb{R}$, then tA (which is defined to be $\{ta : a \in A\}$) is Lebesgue measurable.

Proof. Let $\varepsilon > 0$, and let F be a closed subset of A such that $|A \setminus F| < \varepsilon/t$. Then since F and tF are Borel,

$$\begin{aligned} \varepsilon &> |t||A \setminus F| \\ &= |t|(|A| - |F|) \\ &= |tA| - |tF| \\ &= |tA \setminus tF|. \end{aligned}$$

Since we can find such a closed tF for any $\varepsilon > 0$, tA is Lebesgue measurable.

□